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Development of Power-Efficient 4-Bit Arithmetic Logic Unit Using PD-RBB

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***Abstract*— The Arithmetic Logic Unit (ALU) is a critical component in digital circuits, responsible for executing arithmetic and logical operations. Hence the performance enhancement of digital circuits with a power-efficient ALU is very much important. This paper focuses on Design and development of an optimized 4-bit ALU using Pareto Distribution Reverse Body Biasing (PD-RBB) method. Here, by considering the minimum power loss, the delay is rejected. Also, the photoresist thickness fluctuations are detected by employing Bernstein Polynomial Sigmoid Fuzzy Inference System (BPS-FIS). If photoresist thickness fluctuation is present, then it is mitigated based on PD-RBB. Similarly, by using BPS-FIS, leakage current and transistor failure are detected. The Sigmoid-based Correction Non-Restoring Division (SbC-NRD) is used to perform the complex division modules in ALU operations. The Carry Look Ahead Adder (CLAA) is used to facilitate multiple operations within the ALU. Results shows that, the proposed method achieved a high leakage current reduction of 92% and a sensitivity of 98mV/V.**

Keywords*— Arithmetic Logic Unit (ALU), Carry Look Ahead Adder (CLAA), Division Module Regulation

I. INTRODUCTION

The proliferation of handheld devices and the continuous improvement of digital circuits, particularly the Arithmetic and Logic Unit (ALU), have revolutionized the field of computing and electronics. ALUs serve as the computational core of processors, performing a wide range of arithmetic and logical operations that are fundamental to digital systems. Their versatility and efficiency have made them indispensable in numerous applications beyond traditional computing, such as image and signal processing, where complex mathematical operations are required in real-time.

Despite the advancements in ALU design and functionality, the challenge of developing embedded processors with low power consumption persists. This issue is particularly critical in the context of portable devices, where battery life and heat dissipation are significant concerns. Researchers and engineers are continuously exploring innovative architectures, materials, and design techniques to optimize ALU performance while minimizing power usage. These efforts are crucial for the future development of more efficient and capable handheld devices, as well as for advancing the capabilities of embedded systems in various fields, including IoT devices, wearable technology, and autonomous systems [1-4, 17-20].

Advancement in Digital technology leads to increased area, power consumption, and delay. Consequently, the designs of the computational unit with power-efficient ALU also become biggest challenge. The increasing market demand for low-power, faster, and more compact devices is a key factor in designing devices that satisfy the growing market demand[5-8]. Therefore, developing a 4-bit ALU with both low-power and high-performance is crucial in modern digital circuit design, particularly for applications that prioritize energy efficiency. The difficulty in designing such ALUs arises from the trade-offs between speed, power, and area. Recently many techniques have been developed for designing a low-power with outstanding performance ALU. A 16-bit ripple carry adder/subtractor with XOR gates, a 16-bit pipelined vedic multiplier, and a 16-bit logic unit using operand isolation were employed for designing power-efficient ALU in existing [9-12].

The proposed ALU design incorporates advanced power-saving techniques to minimize energy consumption while maintaining high performance. Simulation results demonstrate significant improvements in power efficiency compared to conventional ALU designs. Future work could explore further optimizations and integration of this power-efficient ALU into larger system-on-chip designs for various low-power applications [13-16].

This remainder of the paper is arranged as follows. Section 2 literature survey is discussed. proposed methodology is conveyed in Section 3. The results and discussion are given in Section 4. Conclusion of the proposed work is given in section 5.

II. RELATED WORK

In [1] framework named Design and Simulation of Reversible ALU. Here, for designing the reversible ALU, the Quantum-dot Cellular Automata (QCA) technology was utilized. Here, reversible BS1 blocks were deployed by the reversible ALU. Therefore, the model excellently improved the quantum cost, occupied area, and the number of cells. Yet, it didn't prevent the risk of transistor failures and soft errors, thus reducing the system's resilience.

In [2] machine learning based Design Technology Co-Optimization (DTCO) for circuits. A ridge regression technique was employed for bypassing the simulation, compact model extraction, and cell library characterization. The model excellently reduced the time consumption, errors of delay, internal power consumption, along with leakage power. However, the research had an irreversibility issue, which might lead to significant input information loss.

In [3] implementation of generic Vedic ALU. The Vedic mathematics principles were utilized for deriving excellent computational methods. Likewise, in the ALU, minimal

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power dissipation and reduced heat generation were achieved by utilizing reversible logic. The model excellently performed fundamental arithmetic operations and bitwise logical operations and also improved the overall performance. The delay arose in the sequential execution of arithmetic and logic operations due to the intricate interconnections among different circuit elements.

In [4] design of Inter of Things (IoT)-enabled arithmetic logic and compression unit was discussed. They integrated compression module and a fast multiplier module into the design of the 16-bit ALU according to the Vedic algorithm. The ALU was synthesized under 32-nanometer High-Voltage Threshold (HVT) cells from the Synopsys database for creating the overview of logic levels and areal efficiency. The model aided in quick computational processes. However, special considerations were not laid on the division module, leading to slower computation and bottleneck issues.

In [5] the design of arithmetic and logical unit was discussed. They used Micro-Ring Resonator (MRR) for performing one-bit addition and one-bit comparison, which made the model easier to integrate with Very Large Scale Integrated Optics (VLSIO). Here, for attaining low-intensity losses, the beam splitter was replaced with MRR. Therefore, an optimized design of ALU was obtained. Nevertheless, the model was sensitive to temperature fluctuations, leading to computation errors.

In [6] ALU for ultra-fast computing based method was discussed. They used minimal fan in/fan out, optimal latency, and high-efficiency ALU module, the model employed superconductor technology and bit-parallel architecture. Likewise, by employing a standard commercial process, the ALU was fabricated. Therefore, the model achieved impressive clock frequency and reduced power consumption. Besides, it superiorly performed arithmetic and logic operations. However, the research introduced additional design and power delivery challenges.

In [7] developed a model for 16-bit ALU with low power. Here, the ALU module was designed based on the Metal–Oxide-Semiconductor Field Effect Transistor (MOSFET) and Fin Field-Effect Transistor (FinFET) technologies. Likewise, the ALU module incorporated the adiabatic switching principle. Here, adiabatic logic families, including Quasi-Static Energy Recovery Logic (QSERL), two Phase drive Adiabatic Dynamic Complementary Metal Oxide Semiconductor (CMOS) Logic (2PADCL), and Adiabatic Dynamic CMOS Logic (ADCL) were used. The model excellently saved the power. Yet, adiabatic logic circuits increased the transistor count, leading to a larger chip area.

In [8] a model for a reversible ALU developed. Hey used reversible gate (i.e., modified 3:3 TSG gate) with QCA technology for optimally designing the ALU with low delay and power dissipation. Therefore, the model reduced the occupied area, delay, complexity, and power. But, the model was prone to fabrication defects and charge leakage, thus introducing errors in computation.

III. Proposed Work

In recent decades, the surge in handheld devices and the advancement of efficient digital circuits like the Arithmetic and Logic Unit (ALU) have captured the interest of researchers. The ALU is responsible for executing arithmetic operations such as addition, subtraction, multiplication, and division, as well as logical operations like OR, AND, XOR, and inversion. Typically, the ALU processes two input operands (A and B) based on predefined operations to generate the output. ALUs are utilized in various applications, including image processing and digital signal processing. Nonetheless, the development of embedded processors with low power consumption remains a significant challenge in all these applications [1-4].

A. Input Operands

Initially, for performing the ALU operation, two input operands are given as input. The photoresist thickness fluctuation is detected while processing these operands, which is explained in the upcoming section. It is expressed as equation (1)

$$\aleph \xrightarrow{operands} (\alpha, \gamma) \tag{1}$$

Here, states the input and signifies the two input operands.

B. Proposed Photoresist Thickness Fluctuation-Aware 4-bit ALU

To detect the photoresist thickness fluctuation, leakage current, and transistor failure, the proposed BPS-FIS is introduced. The proposed PD-RBB is employed for mitigating the photoresist thickness fluctuation and reducing the static power. Likewise, to mitigate the transistor failure, the proposed Wd-TMR is established

The proposed methodology excellently mitigated the photoresist thickness fluctuations, transistor failures, and leakage power for optimal designing of ALU.

1) Photoresist Thickness Fluctuation Detection

Photoresist thickness variations, gate leakage current, power consumption, and thermal instability, the photoresist thickness fluctuation is detected while processing . Bernstein Polynomial Sigmoid Fuzzy Inference System (BPS-FIS) is used for detecting the photoresist thickness fluctuation, thus reducing the gate leakage, power consumption, and thermal instability in ALU operations. Usually, the photoresist thickness variations are not always sharp or binary. The Fuzzy Inference System (FIS) can effectively deal with these gradual deviations. The system may miss the real defects if the membership function is not tuned correctly in FIS. To address this problem, the Bernstein polynomial sigmoid membership function is employed in FIS.

2) Photoresist Thickness Fluctuation Mitigation

If the photoresist thickness fluctuation is present (ϑ), then it is mitigated by using the Pareto Distribution Reverse Body Biasing (PD-RBB) technique. Reverse Body Biasing (RBB) rapidly diminishes the sub threshold leakage by increasing threshold voltage (V_{th}), which is worsened by fabrication variations. Likewise, RBB effectively stabilizes the leakage currents in transistors that are affected by photoresist non-uniformities. But, the high reverse bias may induce undesired body effects in RBB, leading to parasitic leakage paths. To address this problem, Pareto distribution is employed instead of applying a fixed RBB dynamically, thus preventing the parasitic leakage paths.

C. Leakage Current and Transistor Failure Detection

By employing BPS-FIS, leakage current and transistor failure while processing $\aleph$ are detected. The working

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process of BPS-FIS is explained in section A. Here, the fuzzy rule is computed as,

$$\aleph = \begin{cases} if \quad \partial q < 10\mu A \quad Then \quad \lambda \\ if \quad \partial q > 50\mu A \quad Then \quad \varpi \\ if \quad \partial q > 100\mu A \quad Then \quad \eta \end{cases} \tag{2}$$

Here, ∂q states Direct Drain Quiescent Current, μA designates microampere, λ implies the normal range, ϖ signifies the presence of leakage current, and η states the failure in the transistor. Finally, the leakage current and transistor failure detection outcomes (LT) are provided as,

$$LT = \{\lambda, \varpi, \eta\} \tag{3}$$

If leakage current and transistor failure are presented, then it is mitigated, which is explained in the following sections.

D. Static Power Reduction

Static power reduction is performed by utilizing PD-RBB if leakage current is present (ϖ). Here, the leakage current (ϖ) is reduced. The working process of PD-RBB is explained in section 3.3. Static power reduction outcomes are signified as S_n.

E. Division Module Regulation

If any complex division modules $(\wp, \Omega)$ are present while performing ALU operations, then they are regulated based on the Sigmoid-based Correction Non-Restoring Division (SbC-NRD). Usually, the Non-Restoring Division (NRD) effectively reduces the number of operations per cycle to boost the execution speed. An extra correction step is required if the remainder becomes negative at the end, thus increasing computation time. To address this problem, the sigmoid-based correction is utilized in NRD. The steps involved in the SbC-NRD are derived below,

Initially, the quotient (Qt) and remainder (P) are expressed as,

$$Qt = \frac{\wp}{\Omega}, \quad P = \wp \mod \Omega \tag{4}$$

Here, $\wp$ indicates the numerator and Ω specifies the denominator. Next, the (P) is set to zero, and the numerator $(\wp)$ is copied to the quotient register (Qt). For each bit (i.e., from Most Significant Bit (MSB) to Least Significant Bit (LSB)), the left shift and non-restoring subtraction are performed. It is expressed as,

$$\begin{aligned} &P = P << 1 \quad (shift\ by\ 1\ bit) \\ &P = P + MSB \quad of \quad Qt \\ &Qt = Qt << 1 \end{aligned} \tag{5}$$

$$P = P - \Omega \tag{6}$$

Later, the sign of (P) is checked.

$$SC = \begin{cases} if \quad P \geq 0 \quad Then \quad \text{set LSB of } Qt \text{ to 1} \\ if \quad P < 0 \quad Then \quad \text{no restoration} \end{cases} \tag{7}$$

Where, SC designates the sign checking outcomes. Then, for correcting the final remainder $P < 0$, sigmoid-based correction is utilized. It is expressed as,

$$P_{new} = P + \Omega \times sig(P) \tag{8}$$

Here, $sig(P)$ signifies the sigmoid-based correction. Finally, the obtained remainder and quotient are indicated as φ and Ξ.

F. Parallel Processing

Similarly, by using CLAA, parallel processing is done to perform multiple operations simultaneously. CLAA excellently removes the sequential carry propagation delay. The steps involved in the CLAA are explained as follows,

Initially, the generate and propagate for each bit in the (α, γ) are computed. Here, if both bits are 1, then the carry is generated. If at least one bit is 1, then the carry is propagated. It is written as,

$$\begin{aligned} Q_x &= \alpha_x \cdot \gamma_x \\ K_x &= \alpha_x \oplus \gamma_x \end{aligned} \tag{9}$$

Here, x signifies the position of the bit (i.e., 0, 1, 2, 3). Next, all carry values (Y) are estimated rather than waiting for the previous carry. It is formulated as,

$$\begin{aligned} Y_1 &= Q_0 + K_0 Y_0 \\ Y_2 &= Q_1 + K_1 Y_1 \\ Y_3 &= Q_2 + K_2 Y_2 \\ Y_4 &= Q_3 + K_3 Y_3 \end{aligned} \tag{10}$$

The above equation (8) ensures parallel processing, thus making the operation faster. Lastly, the operation is computed. Therefore, an optimal design of ALU is obtained by the proposed methodology.

IV. Results and Discussion

The results demonstrate that the proposed model outperforms existing techniques in terms of accuracy and efficiency. This superior performance can be attributed to the novel approach implemented in the MATLAB environment, which allows for more precise calculations and better handling of complex data. Furthermore, the comparative analysis with RBB, FBB, ABB, and SBC reveals significant improvements in key performance metrics, highlighting the potential of the proposed model for real-world applications.

A. Performance Assessment

Here, the performance of the proposed model is weighed against prevailing techniques to show the dependability of the proposed model.

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The graphical representation of the proposed BPS-FIS and prevailing techniques are depicted in Fig 1. Here, the proposed BPS-FIS took less fuzzification and defuzzification times of 674ms and 723ms, respectively. However, the prevailing techniques, such as Sigmoid FIS, Gaussian FIS took high. Here, to improve the detection of photoresist thickness fluctuations, the Bernstein polynomial sigmoid membership function is modified with FIS.

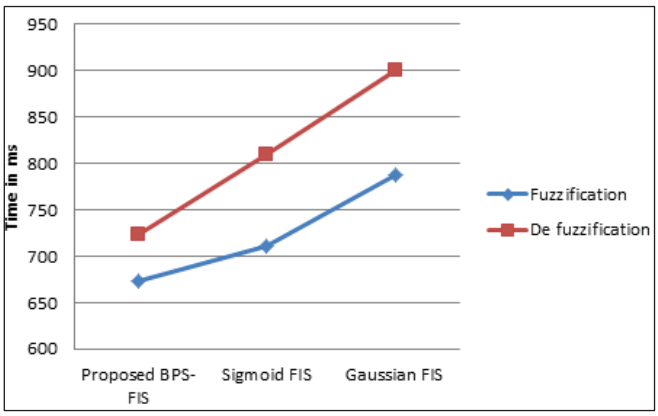


Fig. 1. Graphical representation regarding fuzzification and defuzzification time

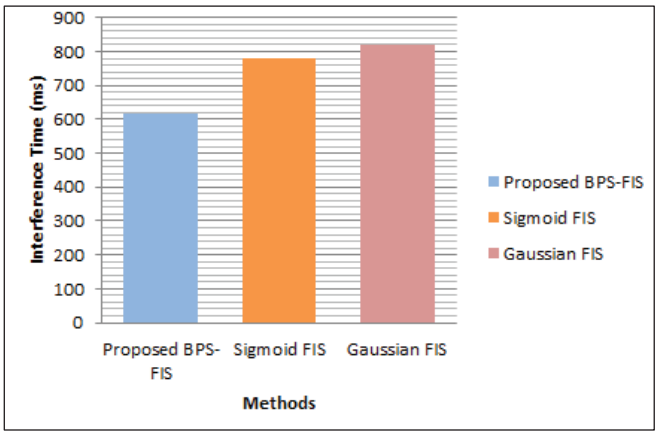


Fig. 2. Inference time analysis

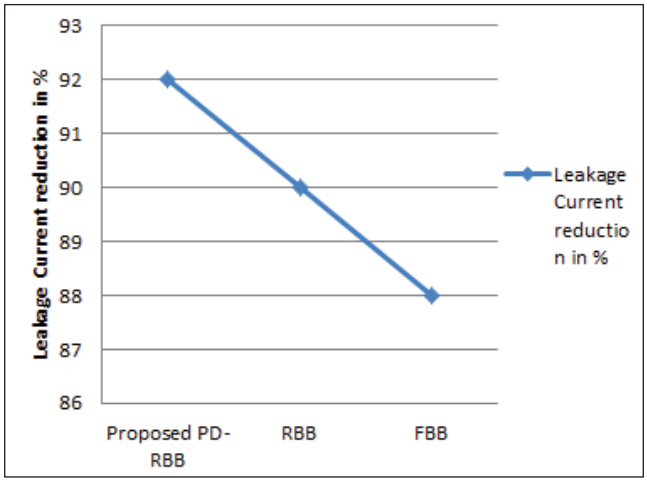
In Fig 2, inference time analysis of the proposed BPS-FIS and conventional methods is shown. Here, for improving the performance of transistor failure and leakage current detection, the BPS-FIS employs the Bernstein polynomial sigmoid membership function. The proposed BPS-FIS took a less inference time of 619ms, whereas the conventional Sigmoid FIS and Gaussian FIS took high inference times of 782ms, 821ms accordingly. Therefore, the effectiveness of the proposed model is proved.

TABLE I. MEMORY UTILIZATION EVALUATION

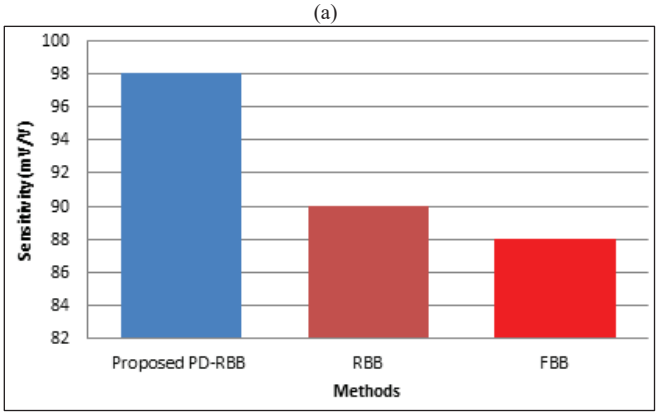
Techniques	Memory Utilization(kB)
Proposed BPS-FIS	391
Sigmoid FIS	510
Gaussian FIS	612

The proposed BPS-FIS uses the Bernstein polynomial sigmoid membership function for improving the identification of leakage current and transistor failure presented while processing the input operands. In Table I, the memory utilization evaluation of the proposed BPS-FIS and prevailing methods is given. Here, low memory of 391kB is deployed by the proposed BPS-FIS. But, the prevailing Sigmoid FIS, Gaussian FIS utilized 510 KB and 612 kB, which was higher than the proposed technique.

B. Performance analysis



(a)



(b)

Fig. 3. Performance validation regarding (a) leakage current reduction and (b) sensitivity

Fig 3 shows the Performance validation regarding leakage current reduction and sensitivity. The performance validation of the proposed PD-RBB and existing techniques regarding leakage current reduction and sensitivity is depicted in Figures 3 (a) and (b). Here, the proposed PD-RBB achieved a high leakage current reduction of 92% and sensitivity of 98 milliVolts per Volt (mV/V) owing to the usage of Pareto distribution. Likewise, the existing RBB, Forward Body Biasing (FBB) obtained low leakage current reduction of 90% and 88% respectively. Likewise, the existing techniques attained low sensitivity. Therefore, the proposed model's reliability is proved.

TABLE II. COMPARATIVE EVALUATION BASED ON PERFORMANCE METRICS

Methods	Sleep State Leakage Reduction (%)	Total Power Consumption (µW)
Proposed PD-RBB	94	458
RBB	90	598
FBB	88	613

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The comparative evaluation of the proposed PD-RBB and conventional techniques with respect to sleep state leakage reduction and total power consumption are depicted in Table II. Here, to prevent the parasitic leakage paths, the proposed PD-RBB employs Pareto distribution. The proposed PD-RBB achieved a high sleep state leakage reduction of 94% and consumed a low total power of 458 microWatt (μW). Yet, the conventional RBB and FBB obtained low sleep state leakage reduction of 90% and 88%, respectively. Therefore, the proposed model is better than the conventional techniques.

C. *Comparative Analysis*

Comparative analysis of the proposed works is provided in Table III.

TABLE III. COMPARATIVE EVALUATION OF THE PROPOSED AND RELATED WORKS

Author's name	Objective	Techniques	Pros	Cons
Proposed Model	Optimal designing of power-efficient 4-bit ALU	PD-RBB	The proposed model excellently detected and mitigated the photoresist thickness fluctuations in design of 4-bit ALU.	However, it failed to consider the defect density correlation with resist thickness fluctuations in the design of ALU.
(Ajitha & Chandra Sekhar Reddy, 2021)	Designing signal-gating aware energy-efficient 8-bit ALU	HFA and ASG	The model consumed less power.	It might increase the overall chip area.
(Zahoor et al., 2021)	Implementing ternary ALU	CNTFET	The research enabled compact circuit realization.	The model increased the design complexity.

The comparative evaluation of the proposed along with related works are depicted in Table III. Here, in the designing of 4-Bit ALU, the proposed PD-RBB superiorly detected and mitigated the photoresist thickness fluctuations. Yet, the overall chip area is augmented by the existing Hybrid Full Adder (HFA) and Adaptive Signal Gating (ASG). Likewise, the conventional Carbon Nanotube Field Effect Transistor (CNTFET) had design complexity. Therefore, the proposed model is better than the existing works.

V. CONCLUSION

In this paper an effective framework named PD-RBB based photoresist thickness fluctuation mitigation-aware optimal designing of power-efficient 4-bit ALU is presented. Important processes, including photoresist thickness fluctuation detection and mitigation, leakage current and transistor failure detection, static power reduction, transistor failure mitigation, irreversibility issue avoidance, delay rejection, division module regulation, and parallel processing were performed in the proposed methodology. Here, for mitigating the photoresist thickness fluctuations, the proposed PD-RBB achieved a high leakage current reduction of 92% and a sensitivity of 98mV/V. Also, for detecting the leakage current and transistor failure, the proposed BPS-FIS took a less inference time of 619ms

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